E – Commerce Analytics Project Report

End-to-End Data Analysis Project

Author: Yusuf Baig
Date: October 2026
Tools: MySQL · Power BI · Power Query · DAX

GitHub: Yusufbaig2001/ecommerce-analytics-olist

1.EXECUTIVE SUMMARY

This project analyzed over 99,000 orders from the Brazilian E-Commerce Public Dataset (Olist) to uncover revenue drivers, customer behaviors, delivery performance and seller quality. Using MySQL for data storage and analysis and Power BI for visualization, we identified 6 key business insights and proposed 5 actionable recommendations. The final deliverable includes a 9-table relational database, 30 business analysis queries, and a 4-page interactive dashboard.

Headline metrics:

- Total Revenue: **$ 16.01 M**
- Total Orders: **99,441**
- Total Customers: **96,096**
- Average Order Value: **$ 160.99**
- Average Review Score: **4.09/5**
- Late Delivery Rate: **7.87%**

2.BUSINESS PROBLEM

An e-commerce marketplace needed to understand:

- **Where revenue comes from –** which categories, sellers and regions drive sales
- **How customers behave –** retention, geography, repeat purchase patterns
- **Where operations fail -** delivery delays and seller quality issues
- **What actions to prioritize-** where to invest, what to fix

Without this analysis, decision on marketing spend, logistics partnerships and seller management would be based on guesswork.

3.DATA & METHODOLOGY

3.1Dataset

The Brazilian E-Commerce Public Dataset by Olist contains ~ 100,000 orders placed between 2016 and 2018 across multiple marketplaces in Brazil.

9 raw CSV files:

Table	Rows	Purpose
customers		Customer identity and location
orders		Order status and timestamps
order_items		Products in each order
order_payments		Payment methods and values
order_reviews		Customer ratings and comments
Products		Product details and dimensions
Sellers		Seller locations
geolocation		Zip code coordinates
product_category_translation		Portuguese → English category names

Total: ~ 1.5 million rows.

3.2 Methodology – 5 Phases

Phase 1: Data Profiling

Loaded each CSV into pandas and checked:

- Rows and column counts
- Missing Values
- Duplicates
- Data Types

Key Findings:

- geolocation had 261,831 duplicate rows (26 % of table)
- order_reviews had ~ 87k NULL comment titles (customers who only left a score)
- orders had NULL delivery dates (pending/cancelled orders-natural, not errors)

- products had 610 products with missing category names
- All date columns loaded as text (needed conversion)

Phase 2: Database Setup

Designed a 9-table relational schema in MySQL with:

- Primary keys for uniqueness
- Appropriate data types (VARCHAR, INT, DECIMAL, DATETIME)
- Foreign keys to enforce referential integrity

Phase 3: Data Cleaning

- **Deduplication:** Removed 280K duplicate rows from `geolocation`
- **Missing values:** Filled 610 product categories with `'uncategorized'`
- **Median imputation:** Filled 2 missing product weights with the median (700g)
- **Column renames:** Fixed typos (`product_name_lenght` → `product_name_length`)
- **Foreign keys:** Added 6 relationships

Phase 4: SQL Analysis (30 Queries)

Wrote business-oriented SQL queries covering:

- Monthly trends (revenue, orders, AOV)
- Customer analysis (state/city distribution, retention)
- Product analysis (top categories, pricing)
- Seller analysis (volume, ratings, revenue)
- Operational metrics (delivery time, late %)
- Advanced patterns (window functions, CTEs, correlations

Phase 5: Power BI Dashboard (4 Pages)

Built an interactive executive dashboard with:

- KPI cards
- Geographic maps
- Time-series trends
- Category breakdowns
- Repeat vs one-time customer analysis
- Management recommendations

Figure 1: Executive Overview Dashboard

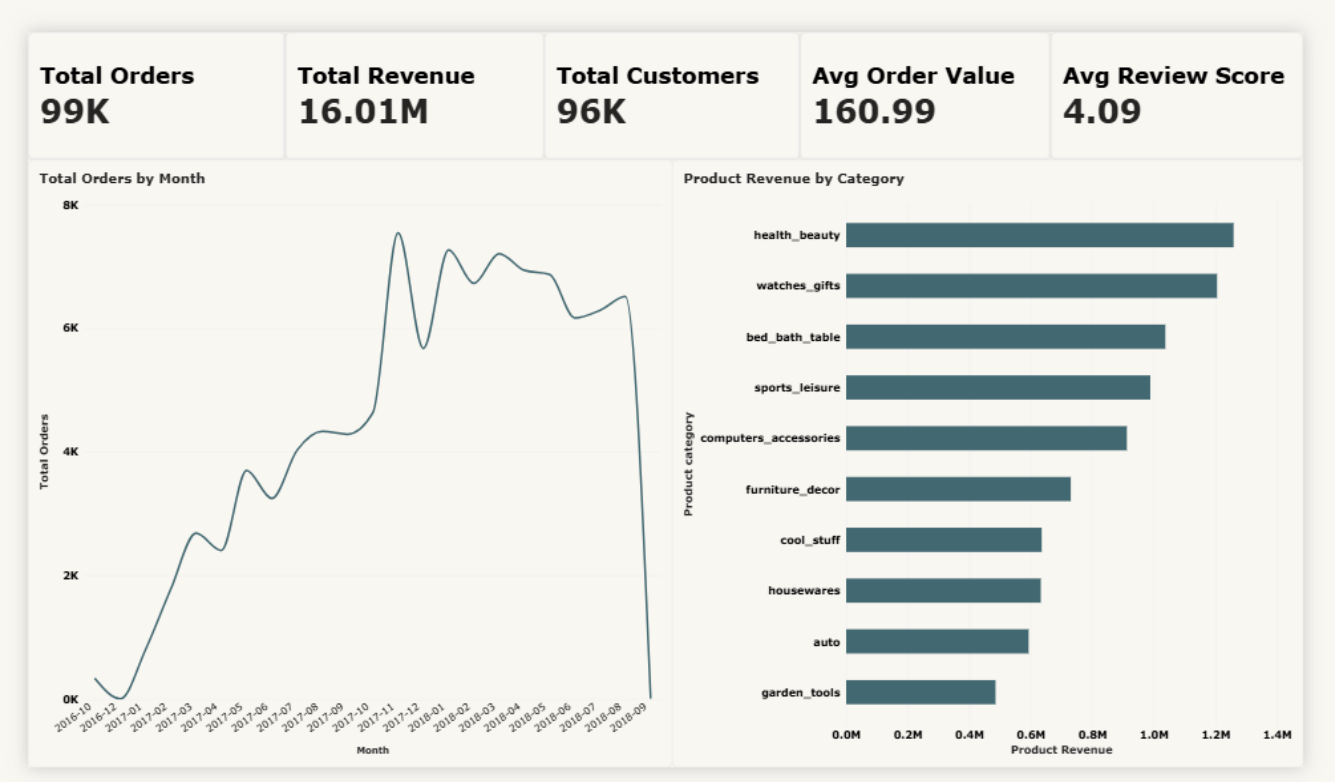


Figure 2: Customer Analytics Dashboard

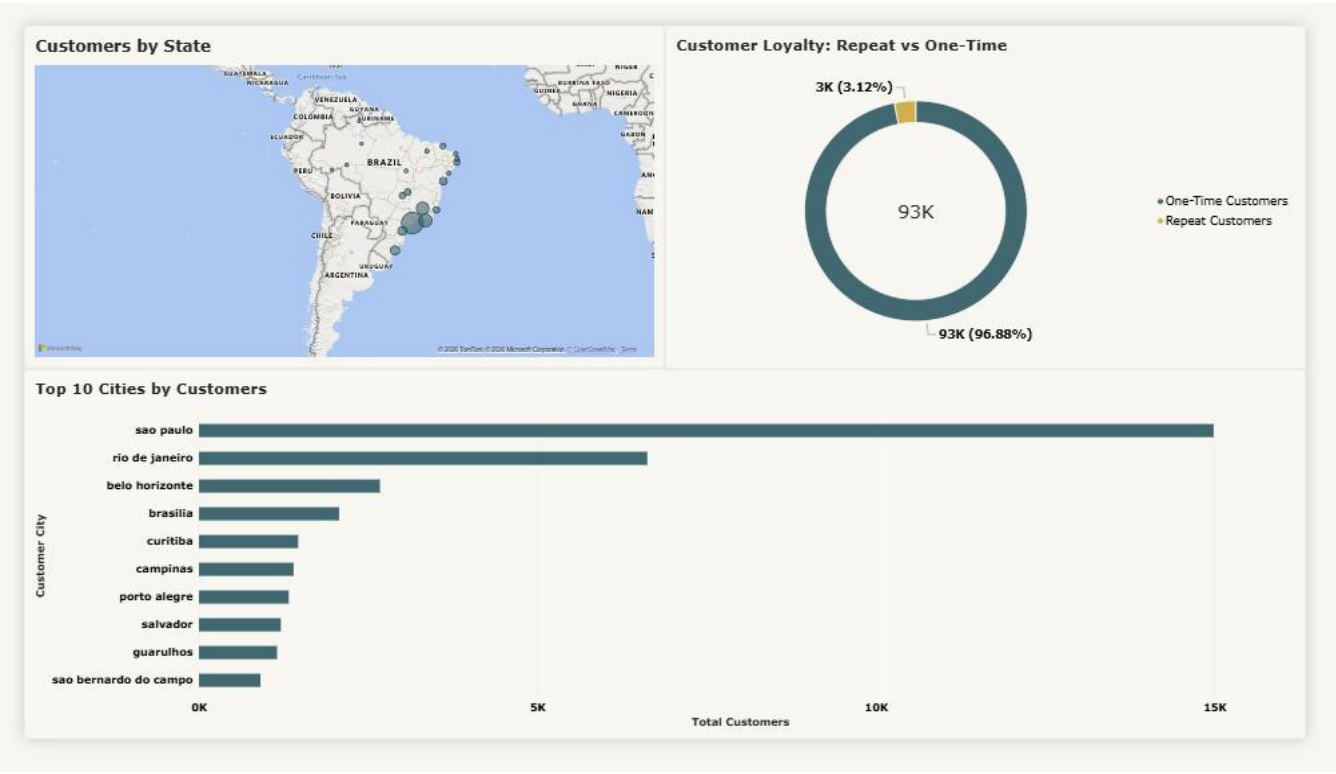


Figure 3: Operations Analytics Dashboard

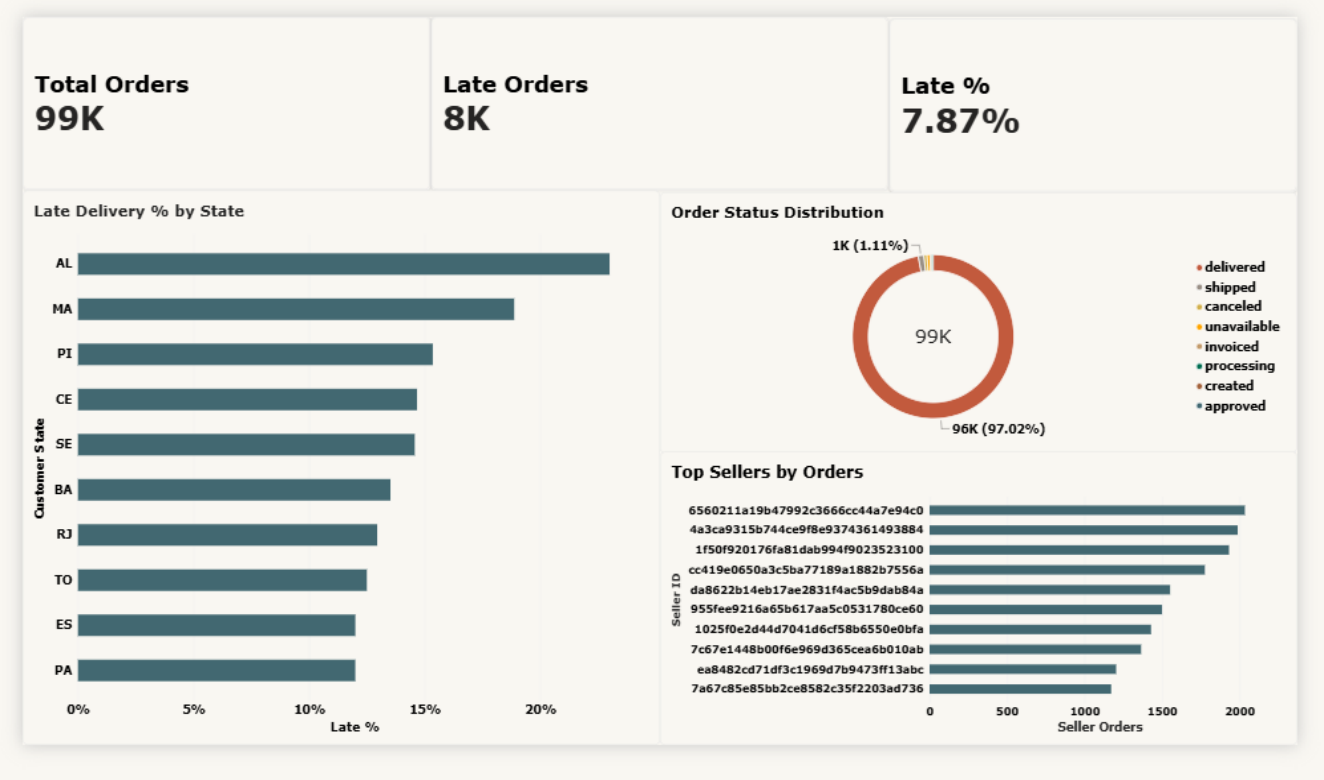
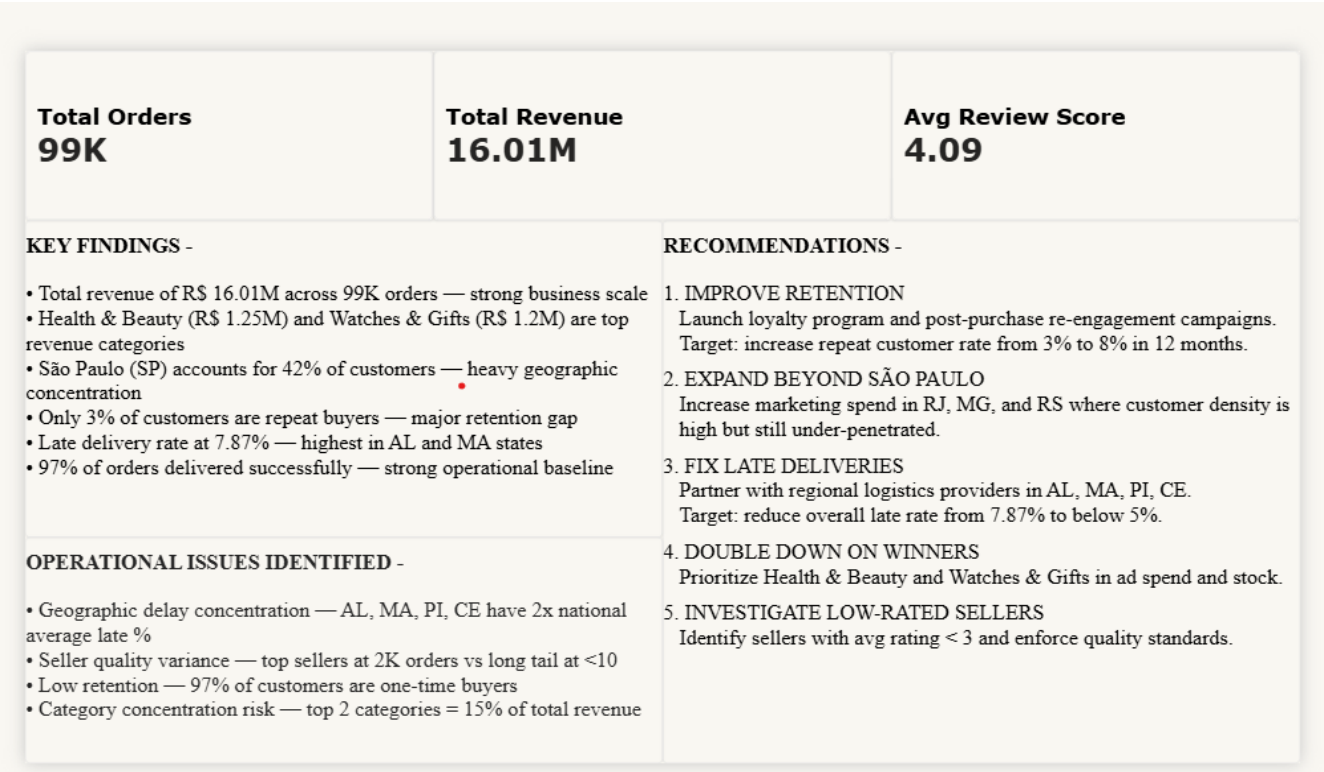


Figure 4: Management Insights Dashboard



4. KEY FINDINGS

4.1 Revenue Drivers

- **Total revenue: $ 16.01M** across 99K orders
- **Top 2 categories** — Health & Beauty (1.25M) and Watches & Gifts (1.2M) — account for **~15% of total revenue**
- **São Paulo (SP) state** generates 42% of all revenue — highly concentrated
- Peak sales months: **November 2017** (Black Friday) and **January 2018**

4.2 Customer Behavior

- **96,096 unique customers**, but only 3% are repeat buyers
- **São Paulo city** alone has 15,540 customers (16% of total)
- Average order value: **$ 160.99**
- Average items per order: **1.14 –** most customers buy a single item

4.3 Product & Seller Insights

- **Watches & Gifts** has the highest revenue per item (avg $ 200+)
- **Top 10 sellers** handle 15% of all orders
- Sellers with 20+ orders average **4.2 stars**; smaller sellers show wider variance
- **71 categories** in English translation

4.4 Operations

- **97% of orders delivered successfully**
- **Late delivery rate: 7.87%** - meaning ~1 in 12 orders arrive after estimated date
- **AL, MA, PI, CE states** have 2x the national average late rate (15-22%)
- **Average delivery time: 12 days** (purchase to door)
- Higher freight costs correlate with **remote northern states**

4.5 Customer Satisfaction

- **Average review score: 4.09/5**
- Categories with **best reviews**: books, food, health & beauty
- Categories with **worst reviews**: fashion, security services
- **Longer delivery → lower review score (negative correlation)**

---

5. RECOMMENDATIONS

1. Improve Customer Retention

- **Problem:** Only 3% of customers are repeat buyers
- **Action:** Launch loyalty program, post-purchase email sequences and personalizes win-back offers
- **Target:** Increase repeat rate from 3% to 8% in 12 months

2. Expand Beyond São Paulo

- **Problem**: 42% concentration in one state creates geographic risk
- **Action**: Increase marketing spend in RJ, MG, RS where density is high but under-penetrated
- **Target**: Reduce SP's revenue share from 42% to 35% in 18 months

3. Fix Late Deliveries in Remote States

- **Problem:** AL, MA, PI, CE average 15-22% late – 2x national average
- **Action:** Partner with regional logistics providers or open regional fulfillment hubs
- **Target:** Reduce national late rate from 7.87% to below 5%

4. Double Down on Winning Categories

- **Action:** Prioritize ad spend and inventory on Health & Beauty and Watches & Gifts
- **Rationale:** These 2 categories = 15% of revenue despite being 2 of 71 categories

5. Investigate Low-Rated Sellers

- **Problem:** Some sellers consistency scores below 3 stars
- **Action:** Enforce quality standards, offer seller training, or delist repeat offenders
- **Target:** Raise floor seller rating from <3 to ≥3.5

---

6. LIMITATIONS

- **Data period:** 2016-2018 – trends may not reflect current market
- **Missing values:** Review comments (~ 87K NULLs) limit text-based sentiment analysis
- **No cost data:** Cannot calculate true profit margins – only revenue
- **No session data:** Cannot analyze pre-purchase behavior (browsing, cart, abandonment)
- **Geographic granularity:** State-level only; city-level would be more actionable

---

7. NEXT STEPS

To extend this analysis:

- **Customer segmentation** using RFM (Recency, Frequency, Monetary)
- **Predictive modeling –** churn prediction using Python (scikit-learn)
- **Time-**series forecasting – forecast next quarter's revenue
- **Text analysis –** sentiment analysis on review comments (NLP)
- **Automated dashboard refresh –** connect Power BI to live database

---

8. TECHNICAL APPENDIX

SQL Techniques Used

- **Joins:** INNER, LEFT across 4+ tables
- **Window Functions:** RANK, DENSE_RANK, ROW_NUMBER, LAG, LEAD
- **CTEs:** Multi-step analytical pipelines
- **Conditional Aggregation:** CASE WHEN inside SUM/COUNT
- **Date Functions:** DATE_FORMAT, DATEDIFF, TIMESTAMPDIFF, YEAR, MONTH
- **Subqueries:** Scalar and correlated

Power Bi Techniques Used

- **DAX Measures: Total Revenue, AOV, Late %, Repeat Customers, Customer Tier**
- **Power Query: Data type fixes, trimming, removing blanks**
- **Data Modeling: Star schema with 6 relationships**
- **Visuals: KPI cards, line charts, bar charts, maps, donuts, tables**

Project Folder Structure

```
ecommerce-analytics-olist/
├── SQL/
│   └── analysis_queries.sql # Schema + 30 business queries
├── powerbi_dashboard/
│   ├── page1_executive.png # Executive Overview
│   ├── page2_customers.png # Customer Analytics
│   ├── page3_operations.png # Operations Analytics
│   ├── page4_insights.png # Management Insights
│   └── README.md # Dashboard documentation
├── Documentation/
│   └── Project_Report.pdf # This report
└── README.md # Project overview
```